Absence of reflection-symmetric two-periodic boundary laws in the competing three-state SOS model

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Abstract

We consider the three-state solid-on-solid model on a rooted binary tree with nearest-neighbor and sibling interactions, each sibling bond counted once. For arbitrary finite real couplings and positive finite inverse temperature, we prove that the exact recursion of positive reflection-symmetric boundary messages has no nonfixed two-cycle. This resolves the two-periodicity conjecture of Karshiboev and Rahmatullaev in its stated invariant subspace. The proof reduces two distinct cycle values to their sum and product. Positivity would force a negative linear coefficient, but in that entire parameter region a strict inequality makes the corresponding discriminant negative. We also give a square-plus-positive-remainder certificate for the composition quotient. The boundary-message kernels yield compatible Gibbs cylinder distributions, so the result excludes alternating generation-parity boundary laws in this class. It does not assert uniqueness or exclude Gibbs measures outside the reflection-symmetric, level-homogeneous class.

Keywords: solid-on-solid model; binary tree; competing interactions; Gibbs measures; periodic boundary laws

1 Introduction

Periodic boundary messages provide a precise way to test for alternating equilibrium states on a rooted tree. In the three-state solid-on-solid (SOS) model with an additional interaction between siblings, the two children can remain correlated after conditioning on their parent. Nevertheless, summing over each parent–children unit gives an exact finite-dimensional recursion. Its reflection-symmetric subspace reduces the question of generation-parity alternation to a scalar two-cycle equation.

Karshiboev and Rahmatullaev derived this recursion and excluded nonfixed two-cycles in a parameter region described by a nonnegative polynomial coefficient B [1, Sec. 5, Theorem 3]. Their Conjecture 1 leaves the complementary region $B < 0$ to be excluded. Fixed points are omitted from the two-periodic solutions under discussion, as specified immediately after their Eq. (16). The invariant set is $v = 1$, where v is the ratio of the boundary weights for spins 2 and 0.

Here we close that remaining region for all positive values of both Boltzmann parameters. The decisive observation is that $B < 0$ forces $t > 1$ and $|B| < t^4 - 1$, whereas the leading and constant coefficients obey $A > t^6$ and $C > t^8$. These inequalities contradict the existence of two real positive cycle values. The proof is exact and requires no parameter scan or limiting approximation. An alternative global positivity certificate is given in Appendix A.

2 Model and boundary messages

Let every vertex of a rooted binary tree have two children, including the root. Write y_x, z_x for the children of x and let $\sigma_x \in \{0, 1, 2\}$. The interaction is

$$\mathcal{H}(\sigma) = -J \sum_{\langle x,y \rangle} |\sigma_x - \sigma_y| - J_1 \sum_x |\sigma_{y_x} - \sigma_{z_x}|. \tag{2.1}$$

The first sum counts every parent–child bond once; the second counts the single unordered sibling pair of each parent once. Infinite-volume Gibbs conditionals use the finite sum of interactions meeting the conditioned volume. We assume finite $J, J_1 \in \mathbb{R}$ and $0 < \beta < \infty$, and set

$$t = e^{\beta J}, \qquad h = e^{\beta J_1}. \tag{2.2}$$

Thus $t, h \in \mathbb{R}_{>0}$ are independent. The couplings have energy units, β has inverse-energy units, and the spin differences and all message ratios are dimensionless. No external field or other distance-two interaction is included. These conventions agree with Karshiboev and Rahmatullaev [1, Eq. (1)], with $(t, h) = (\theta, \theta_1)$.

The Boltzmann factor of a parent of spin i and its labeled children of spins j, k is

$$\psi(i, j, k) = t^{|i-j|+|i-k|} h^{|j-k|}. \tag{2.3}$$

For a positive child message m, define

$$T_i(m) = \sum_{j,k=0}^{2} \psi(i, j, k) m(j) m(k). \tag{2.4}$$

The two assignments (j, k) and (k, j) are distinct when $j \neq k$; this does not double the sibling energy. Descendant subtrees communicate only through the displayed local units, so (2.4) is an exact partition-weight recursion.

Normalize a reflection-symmetric message to $m = (1, u, 1)$ with $u > 0$. Reflection of all three spins gives $T_2 = T_0$. Direct evaluation of the nine summands for each parent spin gives

$$N(u) := T_1(1, u, 1) = u^2 + 4thu + 2t^2(h^2 + 1), \tag{2.5}$$

$$D(u) := T_0(1, u, 1) = t^2u^2 + 2th(t^2 + 1)u + t^4 + 2t^2h^2 + 1. \tag{2.6}$$

Consequently the normalized update is

$$f(u) = \frac{N(u)}{D(u)}, \qquad u > 0. \tag{2.7}$$

Both numerator and denominator are strictly positive. This is the map in Karshiboev and Rahmatullaev [1, Eq. (12), p. 137].

A positive level-homogeneous boundary law means a sequence of messages m_n shared by all vertices in generation n, defined up to positive scalar multiples, with

$$T(m_{n+1}) = \lambda_n m_n, \qquad \lambda_n > 0. \tag{2.8}$$

On the reflection-symmetric subspace this is $u_n = f(u_{n+1})$. An alternating assignment $(1, a, 1), (1, b, 1)$ therefore requires

$$a = f(b), \qquad b = f(a), \qquad a, b > 0. \tag{2.9}$$

A nonfixed two-cycle has $a \neq b$.

3 Exclusion of two-cycles

Theorem 1. *For every $t, h, u > 0$, the function (2.7) satisfies*

$$f(f(u)) = u \quad \Longrightarrow \quad f(u) = u. \tag{3.1}$$

Hence the model (2.1) has no nonconstant generation-parity boundary law among positive reflection-symmetric level-homogeneous messages.

Proof. Write

$$\begin{aligned} N(u) &= u^2 + n_1 u + n_0, \qquad D(u) = t^2 u^2 + d_1 u + d_0, \\ n_1 &= 4th, \quad n_0 = 2t^2(h^2+1), \quad L = d_0 + n_1, \\ d_1 &= 2th(t^2+1), \qquad d_0 = t^4 + 2t^2h^2 + 1. \end{aligned} \tag{3.2}$$

Assume that $a, b > 0$ solve (2.9) with $a \neq b$. Set $F = aD(b) - N(b)$ and $G = bD(a) - N(a)$, so $F = G = 0$. Polynomial expansion yields the two identities

$$F - G = (a-b)(L + a + b - t^2 ab), \tag{3.3}$$

$$F + G + (a+b)(L + a + b - t^2 ab) = 2\big[d_0(a+b) + (d_1+1)ab - n_0\big]. \tag{3.4}$$

Division of (3.3) by $a-b$ is legitimate under the assumption of a nonfixed cycle. With $s = a+b$ and $p = ab$, the two identities become

$$t^2 p = s + L, \qquad d_0 s + (d_1+1)p = n_0. \tag{3.5}$$

Define

$$A = t^2 d_0 + d_1 + 1, \quad B = (d_1+1)L - t^2 n_0, \quad C = d_0 L + n_0. \tag{3.6}$$

Solving (3.5) gives

$$s = -\frac{B}{A}, \qquad p = \frac{C}{A}. \tag{3.7}$$

Here $A > t^6$ since $d_0 > t^4$, and $C > t^8$ since $L > d_0 > t^4$. Moreover,

$$\begin{aligned} B = 1 - t^4 &+ 2th(t^6 + t^4 + t^2 + 3) \\ &+ 2t^2h^2(3t^2+5) + 4t^3h^3(t^2+1) > 1 - t^4. \end{aligned} \tag{3.8}$$

Because $s > 0$ and $A > 0$, equation (3.7) forces $B < 0$. Inequality (3.8) then implies $t > 1$ and $0 < -B < t^4 - 1$. Therefore

$$B^2 < (t^4-1)^2 < t^8 < 4t^{14} < 4AC. \tag{3.9}$$

On the other hand, two real numbers a, b with sum and product (3.7) necessarily satisfy

$$A^2(a-b)^2 = B^2 - 4AC \geq 0, \tag{3.10}$$

contradicting (3.9). No nonfixed positive two-cycle exists. Finally, if $f(f(u)) = u$, positivity of $f(u)$ allows $a = u$, $b = f(u)$ in this argument. They must coincide. □

The coefficients (3.6) define the quadratic obtained by eliminating fixed points from the second iterate of (2.7); compare Karshiboev and Rahmatullaev [1, Eq. (17), p. 139]. When $B \geq 0$, $A, C > 0$ already exclude positive roots. Equation (3.9) excludes the entire remaining region $B < 0$, thus resolving Conjecture 1 in the nonfixed sense of its preceding discussion. This includes the boundary $B = 0$. The statement is not that $B^2 - 4AC$ is negative for all positive t, h; the sign implication is needed only when $B < 0$.

4 Gibbs interpretation and scope

For completeness, the compatibility condition (2.8) has a direct infinite-tree interpretation despite the sibling coupling. For any compatible positive messages, put

$$K_n(j,k \mid i) = \frac{\psi(i,j,k)m_{n+1}(j)m_{n+1}(k)}{\lambda_n m_n(i)}. \tag{4.1}$$

Its sum over j,k is one by (2.8). Give the root spin the law $m_0(i)/\sum_r m_0(r)$, and multiply these conditional pair kernels over the parents in successive generations.

Let W_n denote generation n and $V_n = \bigcup_{k=0}^{n} W_k$. The messages at internal vertices cancel, leaving the cylinder density

$$\mu_n(\sigma_{V_n}) = \frac{1}{Z_n} \prod_{x \in V_{n-1}} \psi(\sigma_x, \sigma_{y_x}, \sigma_{z_x}) \prod_{x \in W_n} m_n(\sigma_x), \tag{4.2}$$

where $Z_n > 0$ normalizes the density. Summing a terminal generation recovers μ_{n-1} because each kernel is normalized. These consistent distributions on finite spin spaces define an infinite probability measure.

To check the Gibbs condition for a finite vertex set Λ, choose V_n large enough that all local units meeting Λ lie inside it and the terminal generation is outside Λ. Conditioning (4.2) on the spins outside Λ cancels terminal weights and all interactions disjoint from Λ. The remaining factor product is exactly the Boltzmann weight of every interaction meeting Λ, including a sibling bond crossing its boundary. This conditional depends on only finitely many exterior spins; the same identity persists in the consistent infinite measure. Thus these boundary laws give the Gibbs specification of (2.1).

Theorem 1 excludes an alternating assignment in this positive reflection-symmetric class. It does not classify all Gibbs measures or establish uniqueness, extremality, stability, or convergence of arbitrary message iterations. In particular, reflection symmetry and level homogeneity are hypotheses; the theorem does not exclude two-periodic messages with unequal spin-0 and spin-2 weights. This agrees with the scope distinction in Karshiboev and Rahmatullaev [1, Remark 5(a)]. No zero-temperature conclusion is taken.

The nearest-neighbor limit also fixes the message convention. When $h = 1$,

$$f(u) = \left(\frac{u+2t}{tu+t^2+1}\right)^2. \tag{4.3}$$

The Hamiltonian of the later thermodynamic study [2, Eq. (2.1), p. 166] has only nearest-neighbor interactions. After shifting its spin labels by one, its branch recursion is $g_{n+1} = W(g_n^2)$, where $W_{ij} = t^{|i-j|}$ and the square is componentwise [2, Eq. (2.4), p. 166]. Our descendant message satisfies $T(m) = (Wm)^2$ at $h = 1$, so the convention change is $m = g^2$. This one-coupling limit does not cover independent $h \neq 1$; no other thermodynamic or dynamical conclusions of that study are used here.

5 Exact verification

The accompanying code contains two exact representations of the calculation. The integer checker reconstructs all 27 local weights in (2.3), forms two generations of the unnormalized update, and verifies the composition identity in Appendix A by sparse polynomial arithmetic over the integers. It uses no computer algebra library or cached coefficient data. The symbolic checker reconstructs the local update as a quadratic form with sibling matrix $S_{jk} = h^{|j-k|}$ and independently derives (3.3)–(3.7). Both verify the positive remainder in (A.6).

As a finite-volume check of (4.2), the symbolic-check program also uses exact rational arithmetic at $t = 3/2$, $h = 2/5$ and terminal message $(1, 5/7, 1)$. On a seven-vertex tree the kernel product agrees with the direct normalized Hamiltonian weight for all $3^7 = 2187$ configurations. Its first-generation marginals and all $3^6 = 729$ exterior choices for an internal-site conditional retain the crossing sibling interaction. These finite checks test the implementation; the infinite-volume statement follows from the consistency argument above. The algebraic identities hold for all parameters in their stated domain, without floating-point error, truncation, or extrapolation.

A A positive composition quotient

An alternative certificate avoids dividing by a fixed-point factor. With the notation of (3.2) and (3.6), set

$$\begin{aligned} P(u) &= N(u) - uD(u), \qquad Q(u) = Au^2 + Bu + C, \\ E(u) &= t^2 N(u)^2 + d_1 N(u)D(u) + d_0 D(u)^2 > 0. \end{aligned} \tag{A.1}$$

Direct polynomial multiplication gives

$$N^2 + n_1 ND + n_0 D^2 - uE = PQ, \qquad f(f(u)) - u = \frac{P(u)Q(u)}{E(u)}. \tag{A.2}$$

These are identities including at $P = 0$. All coefficients are fixed by the defining map; in particular the expansion of C contains $4t^6h^2$.

Separate the terms of positive degree in h:

$$Q(u) = Q_0(u) + R_h(u), \tag{A.3}$$

$$Q_0(u) = (t^6 + t^2 + 1)u^2 + (1 - t^4)u + t^8 + 2t^4 + 2t^2 + 1, \tag{A.4}$$

$$\begin{aligned} R_h(u) = {} & 4t^4h^4 + h^3(4t^5u + 4t^3u + 8t^3) \\ & + h^2(2t^4u^2 + 6t^4u + 10t^2u + 4t^6 + 6t^2) \\ & + h(2t^3u^2 + 2tu^2 + 2t^7u + 2t^5u + 2t^3u + 6tu + 4t^5 + 4t). \end{aligned} \tag{A.5}$$

Every monomial in R_h is positive for $t, h, u > 0$. Completing the single negative term in Q_0 into a square gives

$$Q(u) = t^2\left(t^2u - \tfrac{1}{2}\right)^2 + (t^2 + 1)u^2 + u + t^8 + 2t^4 + \tfrac{7}{4}t^2 + 1 + R_h(u) > 1. \tag{A.6}$$

Here the formal substitution $h = 0$ used to define Q_0 is only an algebraic device; it does not introduce a zero Boltzmann weight into the physical domain. Equations (A.2) and (A.6) imply $f(f(u)) = u$ if and only if $P(u) = 0$, which gives a second proof of Theorem 1. The certificate covers $t = 1$, $B = 0$, repeated fixed points, and any removable algebraic factors of f without cancellation at a fixed point.

Data availability

All defining mathematical inputs are given in this article. The accompanying reproducibility files contain the integer and symbolic checkers, their environment requirements, and instructions for regenerating the exact outputs. No empirical data or fitted parameters are used.

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OpenAI Codex was used for proof exploration, algebraic reasoning, code generation, source checking, internal computational review, and manuscript drafting. This assistance included substantive work on the derivation. The exact program checks described above do not constitute external peer review. Human critical review and final approval of this draft are pending.

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